

Can AI Agents Analyze Gamma-Ray Bursts?

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ABSTRACT

Yes—today with a human at the gate, and the gate is the training apparatus, not the ceiling. Analyzing a gamma-ray burst (GRB) is mostly judgement calls: which detectors to trust, where the background sits, where the burst starts and stops, how finely to slice it in time, and which of two dozen spectral models to believe. We describe a system that makes those calls, and we are precise about its architecture: it is not an autonomous agent but a *gated workflow with bounded agency*—predefined analysis paths in which an AI makes named, auditable decisions (model selection, fit validity, literature verification, mismatch attribution) while a human approves every step, and every approval is stamped. The work is carried by a roster of single-purpose specialist agents—readers, producers, and verifiers—each specialized by the skill documents it reads and the domain tools it commands, fanned out per burst and reassembled under gates, with one founding axiom: a producer never verifies its own work. The system’s knowledge lives in a *Skill Library*: versioned plain-language procedure documents that any AI agent—or any student—can read and execute. The Library is not documentation; it is the system’s memory, and it learns. Each burst is analyzed blind, reconciled against the published literature, and every mismatch is attributed—our error, the literature’s, or a difference of frames—and distilled into a numbered lesson that ships with an executable test, so that no lesson can be silently forgotten. Applied to a campaign over 106 single-pulse *Fermi*/GBM bursts, the walked campaign has distilled thirty-two such lessons^(p), found and fixed engine defects of its own (one of which—a validity gate whose linear margin silently disqualified soft spectra—would have biased the census toward spurious thermal components), confirmed blind predictions drawn from the literature, and refuted one of its own. The learning curve (lessons per burst) is the paper’s central figure: the campaign trains a pipeline, not model weights, and the trained artifact is readable, testable, and transfers between AI systems. We report where the agent cannot yet be trusted—adjudicating contested verdicts, where panels of AI reviewers prove to be strong *generators* of checkable findings but weak *adjudicators*, their unanimity sometimes an artifact of shared priors rather than independent measurement—and we are explicit that this boundary is a statement about the current point on the learning curve, not about AI in principle: every gate correction, and every validated divergence between the human’s choice and the agent’s, is itself distilled into the Library, so the human’s judgment is being transferred, not merely exercised, and the measurable question is whether the gates go quiet. We release the Skill Library, the engine, the tests, and the per-burst records.

Keywords: Gamma-ray bursts (629) — Astronomy data analysis (1858) — Astronomy software (1855)
— Astrostatistics (1882)

1. INTRODUCTION

The question in our title is usually asked as if it had a yes-or-no answer. It does not, because “AI doing anal-

ysis” names at least two different architectures, and the difference between them decides what the answer can mean. In the first architecture, a *workflow*, the analysis follows predefined paths—the same stages, in the same order, for every burst—and software executes them. In the second, an *agent*, a model directs its own process: it decides what to do next, which tools to call, and when

it is finished (Anthropic 2024). Workflows are auditable and dull; agents are flexible and hard to trust. Most claims that “AI analyzed the data” blur the two.

This paper builds and tests a deliberate hybrid, and names it honestly: a *gated workflow with bounded agency*. The analysis pipeline—data acquisition, detector and background selection, adaptive binning, spectral fitting, temporal analysis, quality control—is a workflow: its stages are fixed, its statistics are chosen in advance, and its outputs are stamped with provenance. Inside that workflow, an AI exercises agency at exactly the points where judgement is required and we can afford to audit it: which of twenty-four spectral models wins a gated comparison, whether a fit is valid or sits on a parameter rail, whether a published value has been verified at its source, and—when our numbers disagree with the literature—whether the fault is ours, theirs, or a difference of analysis frames. A human approves every step before the next one runs. Nothing advances on silence, and no approval is ever fabricated: every decision carries a stamp naming who, or what, made it.

Why is time-resolved GRB spectroscopy the right testbed rather than a convenient one? Because it concentrates, in one measurement problem, the three features that make scientific AI hard to evaluate. First, the decisions have no answer key. Two careful published analyses of the same burst, using overlapping data and defensible methods, can disagree about whether a thermal component exists at all; the literature is a *distribution* over defensible choices, not a ground truth (Siddique et al. 2022; Fana Dirirsa et al. 2019). Second, the instrument matters: responses age, detectors are occulted, backgrounds are fit, and a systematic that is invisible in one regime of burst brightness or hardness appears only when the sample happens to present the other. Third, the stakes are population-level: our science application is a census of spectral shapes across 106 single-pulse bursts selected morphologically (Busby & Lazzati 2024), and a small per-burst bias becomes a population-level artifact. A system that merely *runs* on such data is easy; a system whose outputs deserve trust must show its work at every one of these pressure points.

Our design answer, developed in §3, has one unusual component that we believe is the paper’s main methodological contribution. The system’s procedural knowledge lives in a *Skill Library*: versioned, plain-language documents—one per pipeline stage—that state how the stage is run, what its known failure modes are, and what was learned from every burst that ever taught it something. The Library is executable culture: an AI agent reads the same document a new student would, and both are bound by the same rules. Around the Library

sits the second component: a roster of single-purpose specialist agents—each defined by its role, the skills it reads, and the domain tools it commands (§3.3)—and an operating skeleton of typed states, typed failures, and mechanical gates that makes their fan-out and re-assembly auditable (§3.7). Crucially, the Library *learns* under a discipline we call lessons-as-tests: a lesson enters the Library only as a claim with provenance (which burst, which date, which evidence) *and* an executable test that fails if the lesson is ever violated again. Weights are never updated; the harness is. §4 shows what this training regime produced over the first ten bursts of the campaign, including its learning curve, and §5 develops the verification doctrine—blind-first analysis, frozen predictions, and an independence requirement we state as: agreement between two analyses is evidence only in proportion to how deep their shared primitives go.

We are not the first to ask whether AI can do science end-to-end. Multi-agent systems now generate ideas, run analyses, and draft papers across disciplines (Villaescusa-Navarro et al. 2025; Yamada et al. 2025). Our design differs in direction: where those systems are breadth-first—an agent writes fresh analysis code for each project—ours is depth-first: the agents operate a single, tested, domain-specific engine, and the system’s growth is the accumulation of verified domain lessons rather than of generated artifacts. The comparison is instructive in both directions, and we return to it in §8: the breadth-first systems demonstrate, at scale, exactly the evaluation gap—qualitative review standing in for ground truth—that our gated, decision-level scoring is designed to close.

The paper is organized as follows. §2 describes the sample and data. §3 presents the architecture: the Campaign Protocol, the Skill Library, the specialist roster and its domain tools, the authority split between workflow, agent, and human, the doctrine distilled from its failures, and the operating skeleton that enforces it. §4 presents harness learning and the campaign learning curve. §5 presents the verification doctrine and the adversarial experiments. §6 describes the human-versus-agent benchmark design. §7 reports walkthrough-era results, all provisional^(p) until a frozen production sweep. §8 discusses limits, and §9 concludes. Throughout, walkthrough-era numbers carry the flag^(p); they will be regenerated mechanically when the converged pipeline is frozen and run once over the full sample.

2. SAMPLE AND DATA

[PENDING run: port v1 §2 with the L17 inclusion-gate paragraph added]

3. THE AGENTIC WORKFLOW

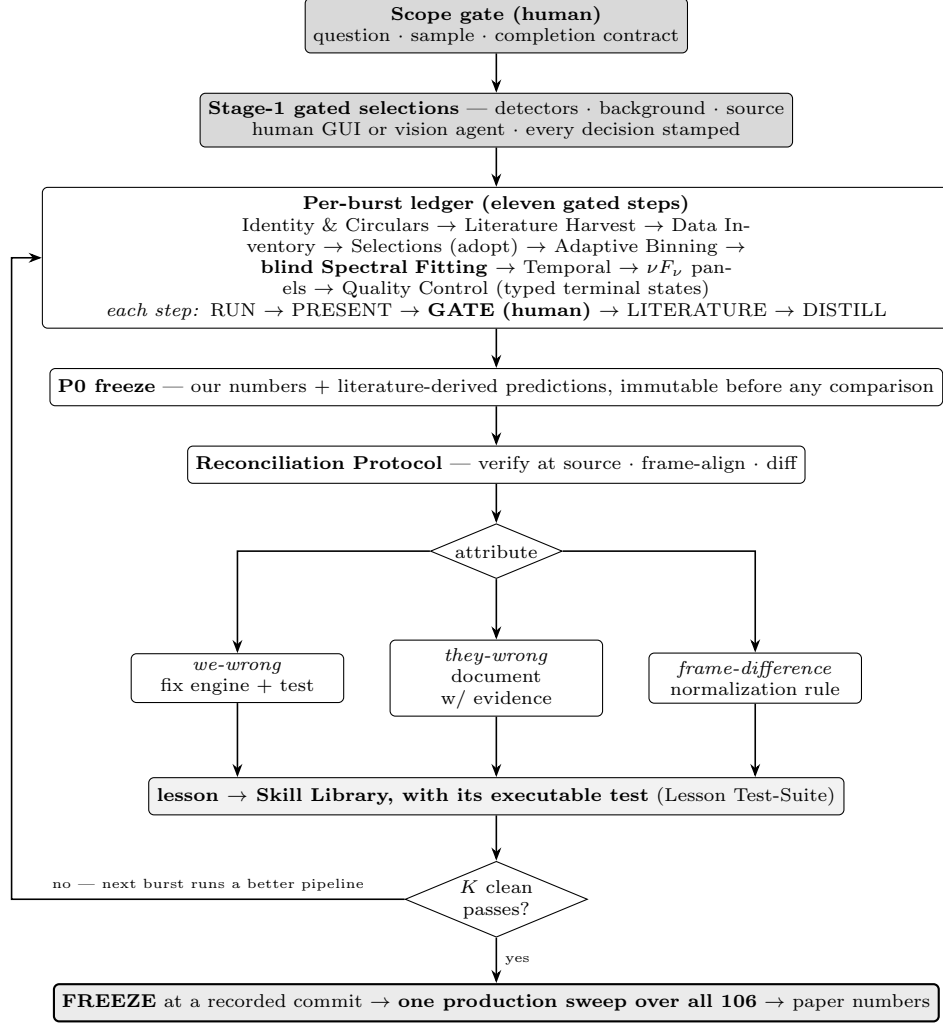


Figure 1. The campaign loop. Humans hold the scope gate, every step gate, lesson acceptance, and the freeze; the agent operates inside the ledger under the Campaign Protocol. The loop’s product is the matured Skill Library; only the frozen production sweep produces citable numbers.

3.1. The campaign loop

The unit of work is one burst, walked through eleven gated steps (Fig. 1). The *Identity & Circulars* step resolves which burst this actually is—same-day triggers exist, and we show below that this step has already intercepted a wrong redshift—and reads every real-time circular. The *Literature Harvest* step finds and fetches every refereed paper that analyzed the burst, using four query forms (name, trigger name, day-fraction, trigger number), preferring the version of record, and recording each harvested value together with its conventions: which peak energy definition, which sign convention, which energy band, which time interval on whose clock, which detectors. The *Data Inventory* step verifies that the downloaded data can support the analysis: file versions, response-matrix time coverage, and detector geometry—including checks that no geometrically well-

placed detector is physically occulted, a failure mode invisible to pointing angles. The *Selection* steps (detectors, background intervals, source interval) are made by a human in a purpose-built interface, or by a vision-capable agent, and in either case are recorded with an approval stamp naming the decision-maker and time; the pipeline treats these stamps as immutable inputs. The *Adaptive Binning* step derives Bayesian-block time bins merged to a significance floor. The *Spectral Fitting* step—where most of the agency lives—is described in §3.2. *Temporal analysis*, *spectral-energy panels*, and *quality control* complete the ledger; the QC step returns one of a set of typed terminal states, of which **non-identifiable** is a *successful* outcome: the workflow is forbidden to keep adjusting an analysis until it manufactures a preference.

Two disciplines wrap the loop. Before any comparison with the literature, the agent’s own results are frozen—

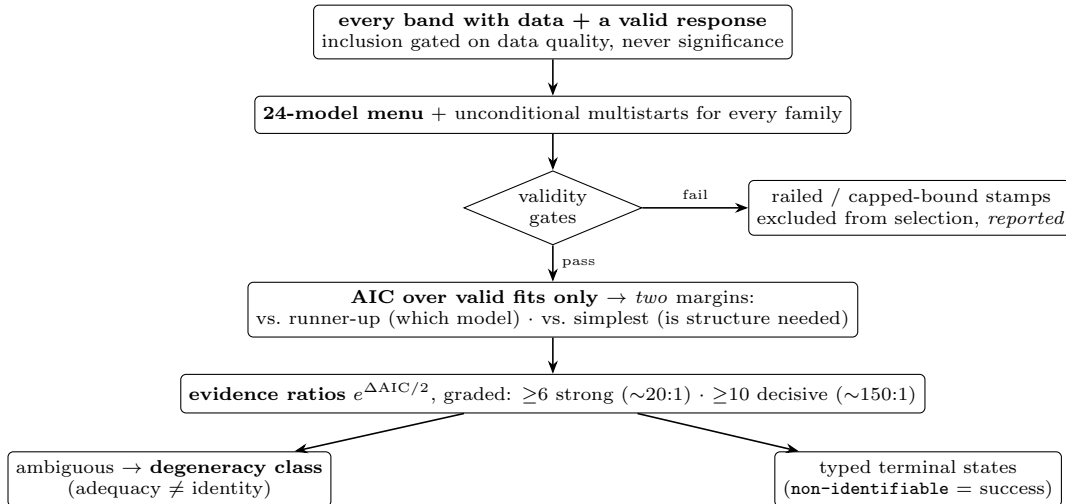


Figure 2. Inside the fitting step. The engine is deterministic; the agent’s bounded agency is the audited interpretation of its outputs.

written to the Burst Record in a form that cannot be revised—together with predictions derived from the harvested literature (§5). Only then are the two compared, and every mismatch is pushed to one of three attributions: *we-wrong* (an error on our side, which must be fixed and tested), *they-wrong* (a documented problem in the published analysis, stated with evidence), or *frame-difference* (both analyses defensible; the difference traced to conventions, data selections, or eras, and recorded as a normalization rule). The attribution, not the mismatch, is the product: it becomes a numbered lesson in the Skill Library, shipped with an executable test (§4).

The loop’s exit condition is convergence: when K consecutive bursts complete without producing a single new lesson, the Skill Library is declared converged, the pipeline is frozen at a recorded commit, and the entire 106-burst sample is re-analyzed once, from scratch, by the frozen system. Only that production sweep produces citable numbers; every value in this paper from the campaign era carries the provisional flag^(p). The design premise, stated plainly: *the per-burst numbers are not the product; the matured Skill Library is.*

3.2. Inside the fitting step

The fitting engine confronts each time bin with twenty-four spectral models: four one-break continua, five carrying extra low-energy structure (thermal or a second break), and fifteen high-energy composites. Three rules govern what the numbers may mean. First, *inclusion is gated on data quality, never on detection significance*: every detector band with usable data and a valid response enters the joint fit, because conditioning inclusion on significance manufactures spurious detections in exactly the bursts where a band fluctuated up-

ward. Second, *every model family receives unconditional multistarts*, because a chained seed from a bright epoch can strand a faint bin in a wrong local minimum—a failure we document in §4. Third, *validity gates* exclude from model selection any fit whose shape parameters sit on a bound, with the margins computed in the parameter’s natural geometry (logarithmic for energy-like parameters spanning decades)—a rule the campaign itself had to learn, at measurable cost, from its first soft burst.

Model comparison is then reported as evidence, not verdicts: the winning model’s margin is quoted both against the runner-up (*which* model) and against the best one-break continuum (*is extra structure needed at all*), each converted to an evidence ratio $e^{\Delta AIC/2}$ and graded. A margin of 6 corresponds to roughly 20:1 and is called strong; 10, roughly 150:1, decisive; anything less is reported as what it is. Where extra structure is required but thermal and non-thermal descriptions sit within a few units of each other, the bin is assigned to a degeneracy class rather than a physics claim: *adequacy and identity are different questions*, and the second frequently cannot be answered from these data alone.

3.3. Specialists: skills, tools, and the fan-out

An “agent” in this system is a small, inspectable thing: a role contract, the skill documents it must read, and the tools it may command. The intelligence is rented from a frozen general model; the specialization is ours, and it lives entirely in that triple. Two readers open every unit of work. A *Skill Reader* begins each step by reading the step’s Skill Library document and defect ledger and returning the binding checklist for *this* burst—the procedure is consumed structurally at every step, never trusted to the operator’s momentum. A *Dis-*

patcher classifies each task by the artifact classes it will produce (figures, numbers, catalog writes, code ports, stochastic products) and returns the roster that must run, in order; a plan in which a producer approves its own output is invalid by construction, and any required guard that does not yet exist is surfaced as declared debt rather than silently relied upon.

Producers then fan out—the fitting engine over time bins, panel builders over models, harvest readers over the published literature—and their outputs converge on verifier gates, each a fresh-context agent with one duty (Table 1): every figure passes a vision check before any human sees it; every printed number is recomputed from the run’s own products; a sub-threshold model-selection margin is reported as a tie, never a winner; a stochastic product must rerun bit-identically from its recorded seed; a ported algorithm is guilty until numerically matched to its source; no row enters a committed catalog unscreened; nothing is re-derived that the project has already proven; and every incident is closed into a lesson, at the correct enforcement layer, in the same session that produced it. A verdict is bound to a content hash of the artifact it certifies and expires the moment that artifact changes.

The tools the producers command are domain instruments, not generic code: the adaptive binning engine (Bayesian blocks merged to a significance floor), the fitting engine of §3.2 with its validity gates, a temporal chain (minimum-variability timescale, cross-correlation lag with its sign convention stated, pulse-shape fits), the νF_ν panel builder, and the literature harvester with its convention ledger. What makes an agent a specialist is not that it can run a tool but that its skill document tells it how to *interpret* the tool: which estimator label rides every timescale, which margin is evidence and which is a tie, which fitted parameter is meaningful only away from its bound, and which published value may be compared only after frame alignment. The assembly is the reverse of the fan-out: verified outputs are composed into the Burst Record, the burst’s typed state advances only when the products on disk evidence it (§3.7), and the record closes against the known-results battery of §5.

3.4. Who decides what

Table 2 states the paper’s honesty contract. The *workflow* decides everything decided in advance: the step order, the energy selections, the statistic, the multistart

seeds, the stamping of provenance. The *AI* decides, within audited bounds: which model wins the gated comparison, whether a fit is valid, whether a published value has been verified at source and in which frame, how a mismatch is attributed, and which anomalies deserve escalation. The *human* decides everything consequential: the scope, every step gate, whether a proposed lesson enters the Library, when convergence is declared, and every scientific claim that leaves the system. In all operation to date^(p), no approval has been generated by the system on the human’s behalf—one approval reached the record seven hours late and arrived typed as late, never fabricated (§7); the reverse—the human overruling the agent’s proposal—occurred repeatedly and is part of the record (§8).

3.5. Two planes

Everything above describes the *production plane*. A second, *discovery plane* exists for ideas that do not belong inside a frozen workflow: anomalies that suggest new observables, hypotheses formed after seeing data. Work there is sandboxed and carries no evidentiary weight until it passes a promotion gate—calibration on simulations, validation on data that did not motivate it, human approval. The governing rule is stated once and enforced always: *the agent may invent a new scorecard after seeing the game; it may not pretend the scorecard was chosen before the game*.

3.6. The doctrine

Seven rules, each purchased by a documented failure, govern all of the above: (1) analyze blind first—published values are harvested before fitting but the fit is never steered toward them; (2) margins are computed over valid fits only; (3) evidence ratios, never bare verdicts; (4) inclusion by data quality, never significance—and its absolute corollary: production fits never drop data that passes the quality gate, with/without refits are attribution diagnostics whose data-dropped member can never be quoted as a result, and omission of data is legal only through a documented quality exclusion; (5) a lesson is not learned until it is a claim *and* a test; (6) no silent approvals—a skipped check is a fake pass; (7) verify claims at their source, and attribute agreement or disagreement at the level of shared primitives. The failures that funded each rule appear in §4 and §8.

3.7. The operating skeleton

The doctrine is enforced by machinery, not memory. Each burst occupies one of fourteen typed states^(p)—from ingested through binned, fitted, and assembled to closed, plus one terminal failure state—and the state is

Table 1. The specialist roster: single-purpose agents around the producers.

specialist	position	binding rule
Skill Reader	opens every step	the procedure is read structurally each time, never recalled
Dispatcher	task intake	a producer never schedules its own approval; missing guards are declared debt
Figure Verifier	before any figure reaches a human	no exceptions; verdict bound to the image hash
Numbers Verifier	before any document ships	a number that cannot be recomputed from products does not print
Tie Reporter	any model-selection report	sub-threshold margins are ties; preference is never argmin alone
Seed Auditor	any stochastic product	a rerun must reproduce bit-identically or the product is not a result
Port Verifier	any ported algorithm	a port is guilty until numerically matched to its source
Admission Gate	any catalog write	no unsanitized row enters a committed catalog
Prior-Art Reader	before any re-derivation	what the project has proven is read, not re-derived
Distiller	after any incident	an incident that ends without a lesson is unfinished

Table 2. The authority split: who decides what.

the workflow decides (predefined)	the AI decides (bounded agency)	the human decides (gates)
step order and Protocol	model winner via the gated comparison	scope and completion contract
energy selections; statistic	fit-validity verdicts	every Stage-1 selection (stamped)
multistart seeds; bound gates	literature verification and frame alignment	every step gate
provenance stamping	mismatch attribution; lesson drafting	lesson acceptance; convergence; freeze
typed terminal states	anomaly flagging and escalation	promotion from the discovery plane; every claim

derived from the products on disk at read time, never asserted, so the system cannot claim progress it cannot evidence. Every failure is classified into one of six classes^(p) (transient, structural, contract, ordering, silent, guard), and each class carries a mandated response: hold and resume; label and continue; stop; wait; demote and make permanently loud; or, for a failed checker, fix the checker and re-run everything it ever gated. The rule that generates the taxonomy: *an error message is never a behavior*—printing a warning is not handling a failure.

Institutional memory is a *Requirements Register*: forty numbered requirements^(p) (one retired), each row recording a needed guard or agent, the incident that revealed the need, and whether the guard is deployed or remains declared debt. The Dispatcher sweeps the Register at task intake, so a missing guard is a stated liability rather than a silent gap. Enforcement obeys a hierarchy—code over hook over artifact over agent over prose—and each rule is moved to the strongest layer that can hold it. Three repository-level hooks^(p) currently enforce, in code, that no figure reaches the human without a hash-bound verifier verdict, that raw data cannot be overwritten, and that no producer runs without a dispatch plan.

Finally, each Burst Record carries a live approval ledger, assembled only from stamps and products it can link—it never asserts what it cannot evidence—and approvals are *revocable by machine*: when an upstream decision is amended, a cascade mechanically demotes every downstream approval that depended on it, and reinstatement requires a human ruling with evidence. Both directions have operated in production: the cascade exposed an approval that had been given in conversation but never stamped (recorded with the delay disclosed rather than back-dated), and a demoted binning approval was reinstated only on proof that the binning’s reference detector was untouched by the amendment, its block table byte-identical^(p). A conformance gate for assembled reports has its contract written and its agent pending; consistent with the Register’s own rule, we state it here as declared debt.

4. LEARNING WITHOUT GRADIENTS

Contemporary AI agents are improved in five ways: pretraining on large corpora; supervised fine-tuning on demonstration trajectories; preference optimization against human or model judgments; reinforcement learning on *verifiable* rewards, where a trajectory is rein-

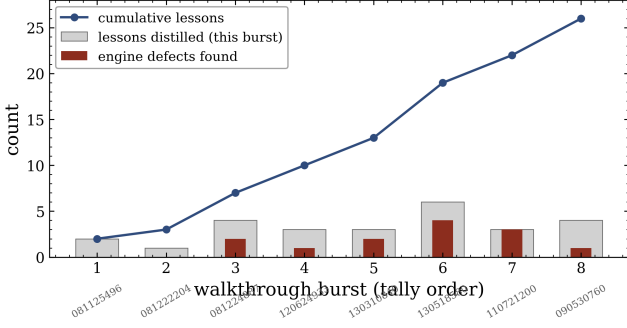


Figure 3. The campaign learning curve^(p): lessons distilled per burst (bars), engine defects found (dark bars), and the cumulative lesson count (line) across the first eight walked bursts. The curve is visibly unconverged; its flattening, not a preset burst count, terminates the training phase.

forced only if its outcome can be checked mechanically (code passing tests, proofs verifying); and *harness-side learning*, in which the model’s weights are never touched and the surrounding harness—its instructions, procedures, and memory—accumulates capability instead (Guo et al. 2025; Wang et al. 2023; Shinn et al. 2023). Our campaign is training of the fifth kind carrying the fourth kind’s essential ingredient. The weights of every model we use are frozen; the artifact that learns is the Skill Library; and the unit of learning is constrained to be *verifiable*: a lesson enters the Library only as a claim with provenance (which burst, which date, which evidence) together with an executable test, added to the Lesson Test-Suite, that fails if the lesson is ever violated again. The suite runs against the current analysis products; where a table predates a fix, the corresponding failure is registered in a known-stale ledger with its clearing condition rather than silenced—debt is visible, never normalized.

This training style has properties that gradient methods cannot offer at our scale, and one property they share. It is data-efficient: thirty-two numbered lessons^(p) from the walked-burst campaign so far. It is auditable: the learned policy is markdown, and every rule cites the burst that paid for it. It transfers across model families: the same Library is executed by different vendors’ agents, which is the basis of the multi-system benchmark of §6. It survives model upgrades. And—the shared property—it exhibits a learning curve (Fig. 3), whose flattening is the campaign’s stopping criterion: when K consecutive bursts produce no new lesson, the Library is converged and the pipeline is frozen.

Two episodes illustrate what the loop learns and how. The first is a defect we designate L27. The engine’s validity gates flagged a fitted parameter as railed when it sat within a fixed *linear* margin of its bound. For

energy-like parameters bounded across three decades, that linear margin is an absolute dead zone of tens of keV above the lower bound—and every burst analyzed in the campaign’s first seven walkthroughs had a peak energy far above it. The eighth burst was soft ($E_p \sim 50\text{--}140 \text{ keV}^{(p)}$), and the gate silently disqualified *every* one-break continuum in most of its time bins, delivering the blocks to thermal-composite models by forfeit. Uncorrected, the census would have manufactured a population-level correlation between spectral softness and thermal components—a selection bias wearing the costume of physics. The repair (margins computed in the parameter’s natural geometry), its unit test, and the burst’s refit happened within the same session; the pre/post contrast—most blocks “preferring” extra structure before, none after^(p)—is the measured size of the bias the lesson prevents. The episode is also an argument about curricula: the defect was invisible until the sample’s serial ordering happened to deliver a soft burst, so *regime coverage, not burst count, is what a training sample must supply*.

The second episode is a cross-validation. Our temporal chain carried a known, ledgered defect: a spectral-lag sign convention inverted with respect to the field’s standard. The eighth burst has a published lag of $+7.08 \pm 0.45 \text{ s}$ in the conventional sign (Lu et al. 2018); our chain returned $-5.99 \text{ s}^{(p)}$. Same burst, same physical direction, opposite sign: the ledgered inversion was confirmed against external ground truth in the course of ordinary operations, and the lesson (state the convention with every quoted lag; validate the sign against a burst with a published direction) was distilled with the defect still unfixed at source—the Library records what is *known*, including known-broken.

4.1. Divergence learning

Failures are not the only teacher. Where the human’s decision and the agent’s proposal diverge and both are defensible, the divergence itself is the signal, and it is processed under a fixed protocol: detect it; elicit the human’s reason *in their own words* (a guessed rationale is forbidden—if the human cannot recall, the record says so); validate the stated reason against the primitives; generalize it only through a measured census over the sample, never from the single case; and enter it in a *Divergence Ledger*. The founding case is a detector-selection divergence: the written geometric rule and two independent agent passes selected seven detectors^(p); the human had selected four^(p). The human’s recollected reason—keep the detectors that triggered the instrument, plus the companion detector on the same spacecraft side—validated exactly against the mission’s trig-

ger record, and reproduces the detector choice of every published analysis of that burst we harvested. A census of the full sample^(p) then measured how much of expert practice that candidate rule explains: about eighty percent^(p), with the remainder requiring judgment the written rule does not capture. The written selection rule therefore stands unamended, and the divergence is recorded as a *measurement of practice*, not promoted to a decision rule. The same census exposed a contract gap—*none* of the 105 human selection records^(p) carried a contemporaneous reason; the single filled record is a retroactive reconstruction and is stamped as such—so rationale capture at decision time is now a registered requirement, while the historical blanks stay blank: a reconstructed reason is never a contemporaneous record. The Ledger gives the paper’s closing question its instrument: the convergence rate—how often the agent’s proposal already matches the human’s decision—is the measurable readiness criterion for ever moving a human gate to an independent approver.

5. VERIFICATION DOCTRINE

5.1. *Blind-first analysis with frozen predictions*

Before each burst is fitted, the Literature Harvest yields a set of falsifiable predictions—derived from published analyses, never from our own outputs—which are frozen alongside a commitment that the fit will not be steered toward them. The comparison happens only after both sides are immutable. The campaign’s scorecards make the point better than argument. For the eighth burst, the literature predicted a positive low-energy photon index early in the burst; our blind fit, with independent binning and a far larger model menu, reproduced it (α up to $+0.6^{(p)}$). It predicted monotonic hard-to-soft evolution; our blind E_p track declined monotonically^(p). But the natural corollary—that a spectrum this hard should demand a thermal component—was *refuted*: with the L27-repaired gates, no time bin required extra structure at even the “strong” evidence grade^(p). A framework that can refute its own frozen expectation is demonstrating the property that matters; confirmation alone demonstrates nothing.

Beyond per-burst predictions, each Burst Record closes against a battery of *known results*: published spectral-lag values with their sign conventions, published Bayesian-block edges, published variability-timescale limits, low-energy photon-index tracks, and—where an instrument we do not fit has published one—polarization constraints entered as external predictions. The battery is how the campaign tests known GRB results and relations as it goes: agreement validates the chain that produced our number, disagreement is pushed

through the attribution of §3.1, and either outcome is banked in the Record.

5.2. *Independence lives at the primitive*

Twice in the campaign, two analyses agreed closely and both were wrong or both unnecessary: our fitted blackbody temperature agreed with a published value to five percent while both components were statistically unrequired; and in the experiment below, two independent review channels built the same load-bearing argument on the same silently symmetric parameterization, corroborating each other into a void conclusion. We therefore state the doctrine as: *agreement between two analyses is evidence only in proportion to the depth of the primitives they do not share*—minimizer, table, column, response model, prior document. N agents reading the same inputs are one opinion wearing N hats.

5.3. *The four-channel experiment: generators, not adjudicators*

We tested whether a panel of AI reviewers could adjudicate a genuinely contested scientific question—a burst for which two published analyses of nearly the same data disagree about a thermal component. Four reviewer channels examined disjoint evidence primitives: the spectral-panel *images* only; the numeric *fit tables* only; the block-to-block *evolution tracks* only; and an *adversary* with full access, briefed to refute the component honestly and permitted to run its own diagnostic refits. A separately-briefed adjudicator then audited all four.

The outcome was double-edged. As *generators*, the channels were excellent: the image channel found a figure-construction defect invisible in any table; the adversary localized a cross-normalization anomaly that we later identified—via the published literature—as a physically occulted detector, and demoted a “decisive” margin to an artifact of a railed reference model, both findings surviving audit and entering the engine as permanent stamps. As *adjudicators*, the channels failed exactly as the doctrine predicts: the two numeric channels derived identical margin ladders and returned opposite verdicts (the disagreement was doctrinal, not evidential); the panel’s one unanimous conclusion was traceable verbatim to a rule in the shared Skill Library—prescribed unanimity, not measurement; and one channel confabulated statistics that do not appear in its own assigned figure, caught only because the adjudicator reopened the figure. Two corollaries entered the doctrine. First, *verify the adversary too*: the adversary claimed a nondeterministic fitting step; five identical reruns showed spread exactly zero^(p)—its variance had been

its own varying invocations. Second, with/without-data refits are attribution diagnostics only; a data-dropped fit is never evidence for a claim (§3.6).

6. EXPERIMENTAL DESIGN: HUMAN VERSUS AGENT

The benchmark design retains v1’s core: a 25-burst subset receives independent Stage-1 selections from experienced researchers, and agents from multiple vendors—executing the *same* Skill Library, so that differences isolate the model rather than the prompt—are scored not on matching any single expert but on landing within the *spread of the experts*, decision by decision: detector sets, background windows, source intervals, model verdicts. Scoring decisions rather than published numbers is forced on us by the field itself: published analyses of the same burst disagree, so the literature is a distribution over defensible choices and cannot serve as an answer key. **[PENDING run: the multi-system benchmark run; a second independent expert completes the human spread]**

We add two further controls in operations. The walkthrough queue runs in two lanes—an older-era lane and a recent-era lane (2021–2025)—so that any skill, constant, or contract that works in one era and fails in the other is exposed as a cross-era defect rather than absorbed as noise. In parallel, a human collaborator replicates the analysis with the Stage-1 selections held fixed, isolating the downstream methodology from selection scatter.

One scoring rule deserves its own statement, because the data forced it. For decisions with *no answer key*, agreement between arms is a *concordance* measure and must never be reported as accuracy: neither arm can be wrong, a divergence is not an error, and any benchmark number that scores it as one is invalid. Background-interval placement is the proving case. For one burst that triggered about two minutes^(p) after a radiation-belt exit, two equally defensible uncontaminated baselines imply excesses in the same interval that differ by a factor of seven^(p) (+68 versus +481 counts s^{−1}): the excess is real while its magnitude is not a measurement, and the published literature uses this very burst as its showcase that polynomial background choices can be ambiguous (Biltzinger et al. 2020). The benchmark therefore separates its dimensions: rule compliance and downstream impact are truth-bearing and scored; window agreement is reported as concordance only. The separation expires only if a physical background model replaces the polynomial one, creating the answer key that does not currently exist.

7. WALKTHROUGH-ERA RESULTS

All numbers in this section are walkthrough-era and provisional^(p); they will be regenerated by the frozen production sweep. The full 106-burst sample has passed once through the fitting engine, with one burst excluded for response-matrix time coverage and reported as an exclusion rather than silently dropped^(p). No cross-burst census sentence is permitted yet: a ledgered retry debt concentrated in the multi-break and composite model families must reach a terminal state first, and per the doctrine that debt is visible in the products, never normalized away. Among the walked bursts: a burst whose apparent two-break “decisive” margin traced to a railed reference model and now carries a capped-bound stamp; a burst whose published photospheric identification our wider menu reduces to a forced choice between statistically equivalent thermal and non-thermal descriptions; a burst in which the fitted low-energy break and the fitted blackbody peak track each other at the few-percent level across eleven bins—one spectral feature wearing two model names; the campaign’s first soft burst (the L27 episode), whose positive low-energy slopes are carried entirely by one-break continua; the campaign’s first intensity-tracking burst, whose blind E_p track follows the flux and whose late-time cutoff evolution sits in open, documented tension with a published constant- Γ curvature interpretation; and the field’s best-known polarimetric burst, for which published polarization-angle rotations enter the frozen predictions as constraints from an instrument we do not fit. The architecture itself is under a live acceptance test: the entire law—skills, roster, gates, skeleton—was handed to a fresh session with no inherited conversational context, on the argument that an architecture that works only for its authors is not yet an architecture. In that arm’s first burst, the first seven of the eleven gated steps have passed their human gates with every approval stamped^(p)—each stamp carrying its own provenance, including one approval given colloquially and typed as read-as-approval, and one recorded late, the case below—and the mechanisms, not the operators’ memories, caught what needed catching. The approval ledger’s cascade exposed an approval that existed only in conversation for seven hours^(p) and forced it onto the record with the delay disclosed. The data-inventory contract surfaced a hundred-millisecond telemetry stall^(p) invisible to the standard exposure accounting (its count deficit is cancelled by a buffered catch-up in the following readout bin), and the repair was accepted only under a robustness criterion: the chosen background cut is the one whose fit quality is independent of the binning grid’s phase, which otherwise moves the fit statistic by up to a factor of 2.4^(p). And a background window found overlapping the approved

source window by about thirty-two seconds^(p) was measured as a real excess ($\geq 3.3\sigma$ in every fit variant^(p)), repaired by an explicit human ruling, and its residual tail contamination carried forward as a disclosed systematic rather than silently accepted. Each catch was typed to its failure class and distilled; the steps beyond the stamped gates remain open and contribute no accepted numbers here.

The Campaign Ledger records, per burst: lessons born, defects found, blind-prediction outcomes, and human gate corrections; it is released with the paper, and Fig. 3 is computed from it. External, zero-cost validations accumulated along the way: our adaptive binning reproduces a published analysis’ Bayesian-block edges to three decimal places on a shared burst (Li & Zhang 2021), and twenty-two of our sample’s bursts carry published lag, width, and evolution-class measurements (Lu et al. 2018) against which the temporal chain is validated burst by burst as the campaign reaches them.

8. DISCUSSION

8.1. Failure taxonomy

Every mode below was observed in this campaign, caught, and converted into doctrine; none was caught by the failing agent itself. *Over-claiming*: three retracted claims, each computed from technically correct numbers whose validity flags had not been checked; caught by the human gate and by an independent agent audit. *Prescribed unanimity*: a panel’s unanimous verdict traced verbatim to the shared prior document (§5). *Confabulation*: a reviewer channel reporting statistics absent from its own assigned figure. *Estimator mismatch*: an ad-hoc quality check using a simpler background model than production, manufacturing a 5.6σ alarm against a selection a human had already, correctly, approved—the rule is that a check must use the production estimator or it is not a check. *Tool-interface drift*: invoking one’s own tools with invented arguments. *Momentum skips*: advancing past a presentation step the Protocol requires; caught by the human, converted into an explicit rule. *Diagnostic-becomes-result drift*: treating a data-dropped sensitivity refit as evidence (§3.6). The taxonomy is the practical answer to “where does the agent still need a human”: at every point where the system’s own priors, momentum, or plausible-but-unchecked numbers are the failure mode, because those are precisely the failures self-review does not catch.

8.2. The capability boundary

The same experiments that expose these failures locate the boundary positively: differently-primed AI reviewers are strong *generators* of checkable findings—

several entered the engine as permanent guards—and weak *adjudicators* of contested verdicts. We therefore spend agency where generation pays (proposing, localizing, refuting, attributing) and reserve adjudication for the gated human, with the audit trail as the interface between them.

8.3. Outlook: the discovery plane, and creativity as graph operations

The production plane described in this paper deliberately excludes open-ended discovery; a second, sandboxed plane exists for it, governed by one rule: the agent may invent a new scorecard after seeing the game, but may not pretend the scorecard was chosen before the game. What would make that plane *generative* rather than merely permitted? We close with a hypothesis we are developing and have begun to stress-test: that creative scientific acts are, predominantly, typed operations on the knowledge graph of the literature—**A'**, drawing an edge along a long, archivally stale, or never-co-cited path between semantically compatible regions; **B'**, negating a *named, load-bearing assumption* of a specific published derivation, as the first generative move rather than as a consequence of new data; and **C**, constructing a new graph element outright—a phenomenon-node, a question, a topology of existing components. An adversarial audit of this taxonomy against twenty-four historically recognized discoveries found the two-class version (A', B') insufficient—class C is required, and the classification must apply to *acts*, not results (a landmark observation can contain no creative jump)—while identifying a genuinely open architectural slot: a literature-scale knowledge graph in which each result carries its derivation assumptions as first-class meta-data, with negation as a provenance-tracked generative operator (Boden 2004; Swanson 1986; Uzzi et al. 2013; Foster et al. 2013; Wiggins 2006). The Reconciliation Protocol already extracts, burst by burst, exactly the assumption-level statements such a graph requires; the discovery plane is where they will compound.

8.4. What generalizes

Gates with stamps, lessons-as-tests, blind-first frozen predictions, frame-alignment before any diff, independence-at-the-primitive, and the failure taxonomy are domain-agnostic and export as a package; the engine, the menu, and the lessons’ content are the domain deposit that makes the package worth executing. Both are released. The package is, in the vocabulary now common for such systems, a *domain harness*: the Library, the Register, the roster contracts, and the doctrine are text and transfer unchanged across vendor

agent platforms, while the enforcement layer—hooks, state derivation, cascades—is rebuilt per platform. We treat that portability as an experiment rather than an assumption: the same law executed by a second vendor’s agent on a known burst, compared gate by gate, is the designed test, and it is pending.

9. CONCLUSIONS

Yes—as a gated workflow with bounded agency, trained harness-side under verifiable lessons, and audited by a doctrine that treats agreement with suspicion and refutation as a service. The system walks bursts end to end under stamped gates, has taught itself thirty-two durable lessons^(p), found its own engine defects including one census-grade selection bias, validated a known defect against published ground truth, refuted one of its own frozen predictions, and now runs as written in a fresh session with no inherited context—the acceptance test of §7. The per-burst numbers are provisional by design; the Skill Library is not. When the learning curve flattens, the pipeline freezes and runs the full sample once; that sweep, not this paper, produces the census. The question in the title has an honest answer only with

its architecture attached—and that architecture, we argue, is the contribution.

We do not mistake the current boundary for a permanent one. Every failure mode in §8.1 was caught at a gate, and every catch became a lesson: the gates are where human judgment is *transferred*, not merely exercised, and the Library is the accumulating transfer. The claim that some capability—taste in ambiguous selections, suspicion of one’s own agreement, the adjudication of contested verdicts—is inherently human is, under this architecture, an empirical claim about the tail of a learning curve, testable by whether the gate-correction rate falls to zero as lessons accumulate. The Divergence Ledger of §4.1 makes that test operational: the convergence rate between the agent’s proposal and the human’s decision is now a defined, recorded quantity, accumulating with every gate^(p). And the curriculum need not stop at analysis: the discovery-plane operators of §8.3—bridging stale regions of the literature graph, negating named load-bearing assumptions, constructing new observables—are mechanisms by which discovery itself becomes teachable. Whether the gates ever go fully quiet is not something this paper can assert; it is something this campaign, continued to convergence and beyond, is built to measure. The gate, in the end, is a curriculum—and curricula are meant to be completed.

APPENDIX

A. REPRODUCIBILITY

The release comprises the Skill Library, the engine, the Lesson Test-Suite, the Campaign Ledger, and the Burst Records, archived with a DOI [**PENDING run: Zenodo concept DOI at release**]. A fresh clone cannot silently reproduce our fits: analysis products are regenerated, not shipped, so reproduction begins at the binning step from tracked inputs; six sample bursts include tracked raw data. The test suite runs in a base scientific Python environment; the fitting engine requires the standard *Fermi* analysis stack.

B. ADVERSARIAL VERIFICATION PROTOCOL

The protocol: independent reviewer channels are assigned disjoint evidence primitives and briefed adversarially; an adjudicator with full access audits every channel’s claims against the primary artifacts; adversary claims are themselves verified by rerun. Results of its first full execution appear in §5. A standing per-burst refuter pass—one honest-refuter agent attacking each Burst Record’s verdict before it closes, its surviving objections recorded in the Record—is the protocol’s next planned extension. [**PENDING run: refuter pass: pending approval and first execution**]

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